B9 Experimental design

Power, sample size and A/B test analysis.

B9-L1 Power and sample size, before fieldwork

Fix baseline, MDE, alpha and power first, compute n, adjust for design effect and subgroups, then field.

```
z_a = NormalDist().inv_cdf(1 - alpha / 2)
z_b = NormalDist().inv_cdf(power)
p_bar = (p1 + p2) / 2
n = ((z_a * sqrt(2 * p_bar * (1 - p_bar)) + z_b * sqrt(p1 * (1 - p1) + p2 * (1 - p2))) / (p2 - p1)) ** 2
```

Four inputs to a sample size	Baseline, minimum detectable effect, alpha, power.
n per group, 40% vs 45%, alpha 0.05, power 0.8	1,534.
Halve the effect you want to detect	About four times the sample.
Weighted or subgroup reads	Multiply n by the design effect. Each subgroup needs the full n.

B9-L2 Analyse an A/B test the way it was randomised

Check the split, keep everyone in their assigned arm, analyse per randomised unit, and report the lift with its interval.

```
from scipy.stats import chisquare
chisquare([50_600, 49_400]).pvalue        # 0.000148, SciPy 1.17
```

```
from statsmodels.stats.proportion import confint_proportions_2indep
confint_proportions_2indep(1150, 10_000, 1000, 10_000)   # (0.0064, 0.0236)
```

First step in any A/B analysis	Sample ratio mismatch check: chi-square on arm counts against the plan.
Intention to treat	Analyse everyone in the arm they were assigned to, whatever happened next.
Unit of analysis	The unit of randomisation, usually the user, not the page view.
Report a lift as	The difference with its interval, compared with the planned minimum effect.

B9-L3 Peeking, novelty and other ways tests mislead

Plan the sample, duration and looks, analyse once at the horizon, and treat anything else as exploration.

Why peeking breaks a test	Each look is another chance for noise to cross $p < 0.05$. Ten looks gave 18.4% false positives in simulation.
Safe alternatives to daily peeking	Fixed horizon, or planned group-sequential looks with stricter thresholds.
Novelty effect	Lift that fades after launch. Run full weeks and compare week 1 with later weeks.
Interference	Treated units affect controls. Randomise at a level that contains the interaction.